Soham Chattopadhyay

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EDUCATION

Bachelor of Electrical Engineering, Honours Kolkata-700032, India | May 2022
JADAVPUR UNIVERSITY

- **Grade** - 9.01/10 (A+)

EXPERIENCE

CVSSP, University of Surrey | COMPUTER VISION RESEARCHER
May 2026 - Present | Prof. Yi-Zhi Song

- Working on **VLM** agents based sketch generation with human feedbacks. The work presents a planner VLM and a drawer VLM trained to plan and iteratively draw sketches, and modify based on human feedbacks.

Addverb Technologies | ENGINEER (COMPUTER VISION & ROBOTICS)
Feb 2025 - Present

- Developed an **rotation-invariant** CNN-based **contrastive autoencoder** to compress high dimensional (**3000**) LiDAR scan to low dimensional latent vector that nearly reduced the scan matching computation by **100 times**. which resulted in faster localization and logging, and reduced the overall **CPU** utilization by **20%**.
- Introduced an end-to-end **RL-based** pallet docking framework for differential drive robots, which utilizes **camera** and **LiDAR** to detect docking stand and an RL policy to generate velocities. The algorithm achieves **5mm** of accuracy.
- Introducing an AI-based software for 'dynamic-obstacles-avoidance' and 'path-planning' in a highly crowded environment. This uses a **camera** for object-detection, a **LiDAR** for **pose estimation** and a deep reinforcement-learning **(RL)** policy for environment-reactive decision making.
- Developing a **multi-sensor fusion** based odometry estimation for mobile robots (AMRs), which includes **CPU-efficient** ground segmentation and moving objects detection in real-time, for eliminating unreliable visual **features**, and a hierarchical fusion of multiple sensors such as **Camera, LiDAR, IMU** and **wheel encoder**.
- Optimized code performance for real-time AruCo markers detection from continuous image-stream, which resulted in an overall **12%** reduction of **CPU utilization**.

Samsung R&D - Delhi | ENGINEER (COMPUTER VISION & ROBOTICS)
Jul 2022 – Jan 2025

- Worked on a research problem to devise **RL-based** adaptive coverage strategy for the Samsung Robot Vacuum Cleaner (RVC). The overall coverage percentage improved by **6%** compared to traditional algorithms.
- Developed the *"Follow-me"* feature for the Ballie, which includes **detection** (Accuracy of **97%** and false positive of **92%**) and **tracking** of various objects using a **Camera** and a **LiDAR**, which runs efficiently on edge-devices.
- Introduced a depth-based probabilistic exploration strategy for frontier exploration to reduce repetition during autonomous mapping. The algorithm reduced repetition by **12%** compared to benchmarking algorithms.
- Devised a Factored Action Reinforcement Learning **(FARL)** based system to find out memory leaks and delay scenarios in UI changes. The RL-agent has been so designed that it can assume multiple actions on a single state making it more robust and efficient compared to traditional RL techniques. The agent reports on an average **120** erroneous scenarios everyday.
- Introduced a continuous learning-based recommendation system for automatic assignment of software issues to their respective owners, which reports an overall accuracy of **93%**.

UiT The Arctic University of Norway, Tromsø, Norway | UNDERGRADUATE RESEARCH INTERN
Jul 2020 – Apr 2021 | Prof. Dilip. K. Prasad

- Developed a novel **1D-CNN**-based deep learning method, called RRCNN, which detects motivation levels of various students by processing EEG signals, with an accuracy of **89%**.
- Developed a deep learning-based novel approach to detect cardiac disease of Salmon Fish at different time stages of their lives. The method effectively learns temporal variation of various features from embryonic heart videos and detects cardiac disease with an accuracy of **72%**.

École de Technologie Supérieure ÉTS, Montréal, Canada | MITACS Global Research Intern'21
June 2021 – Aug 2021 | Prof. Christian Desrosiers

- Developed a weakly-supervised GCN-based segmentation model for brain surface segmentation where the data was mapped to spectral domain.

PATENTS

- *System and Method for Context-Aware and Priority-Driven Projection Area Selection in Multi-user Environment* on behalf of **Samsung R&D - Delhi** [*Link*]

SELECTED PUBLICATIONS

- *'Mapping Functional Changes in the Embryonic Heart of Atlantic Salmon Post Viral Infection Using AI Technique'*, published in **2022 IEEE International Conference on Image Processing (ICIP), IEEE**, publication link : https://ieeexplore.ieee.org/abstract/document/9897203.
- *'Motivation detection using EEG signal analysis by residual-in-residual convolutional neural network'*, published in **Expert Systems With Applications, ELSEVIER**, https://www.sciencedirect.com/science/article/pii/S0957417421009544.
- *'DRDA-Net: Dense residual dual-shuffle attention network for breast cancer classification using histopathological images'*, published in **Computers in Biology and Medicine, ELSEVIER**, publication link : **https://www.sciencedirect.com/science/article/abs/pii/S0010482522002293**.
- *'MTRRE-Net: A deep learning model for detection of breast cancer from histopathological images'*, published in **Computers in Biology and Medicine, ELSEVIER**, publication link : **https://www.sciencedirect.com/science/article/abs/pii/S0010482522008630**.
- *'A feature selection model for speech emotion recognition using clustering-based population generation with hybrid of equilibrium optimizer and atom search optimization algorithm'*, published in **Multimedia Tools and Applications, SPRINGER**, publication link: **https://link.springer.com/article/10.1007/s11042-021-11839-3**
- *'AltWOA: Altruistic Whale Optimization Algorithm for feature selection on microarray datasets'*, published in **Computers in Biology and Medicine, ELSEVIER**, publication link: **https://www.sciencedirect.com/science/article/abs/pii/S001048252200141X**
- *'MRFGRO: a hybrid meta-heuristic feature selection method for screening COVID-19 using deep features'*, published in **Scientific Reports, NATURE**, publication link: **https://www.nature.com/articles/s41598-021-02731-z**
- *'A Hybrid Feature Selection Method using Golden Ratio and Equilibrium Optimization Algorithms for Speech Emotion Recognition'*, published in **IEEE Access**. Publication link : **https://ieeexplore.ieee.org/abstract/document/9247182**
- *'Deep feature selection using beta-Hill Climbing aided whale optimization algorithm for lung and colon cancer detection'*, published in **Biomedical Signal Processing and Control, ELSEVIER**. Publication link : **https://www.sciencedirect.com/science/article/abs/pii/S1746809423001258**

SKILLS

Experties: • Vision Language Model (VLM) • Robot perception • Deep Learning • Embodied AI • Physical AI • Reinforcement Learning • Self-supervised Learning • Contrastive learning • Image processing • CNN • Transformer • ONNX • TensorRT • Tf-lite • Diffusion model • Model Quantization • Physical AI • SLAM / vSLAM • PCL • 3-D Vision • Multi-modal Sensor Fusion • Pose Graph Optimization
Programming Languages : • C/C++ (Advanced) • Python (Advanced)
Engineering : • Edge deployment • Low level architecture design • System engineering • Performance optimization • Clean code
Tools & Frameworks : • Pytorch • Tensorflow • ROS/ROS2 • gRPC • Perf • Ceres • Git/Github/Gitlab • Docker • GDB
Problem solving : • 1400+ problems (1700*) @Leetcode

AWARDS & ACHIEVEMENTS

- MITACS GRI'21
- 2X Spot award receiver @Samsung R&D
- 2X Stellar Project award winner @Samsung R&D
- 'Top finalist' @Samsung R&D India Hackathon

EXTRA-CURRICULAR

- **Teaching**: Take regular sessions on fundamentals of deep learning, transformers and reinforcement learning
- **Interviewer**: Actively take part in talent acquisition by taking technical interview rounds.